Research article

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A Bayesian marked spatial point processes model for basketball shot chart

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Abstract: The success rate of a basketball shot may be higher at locations where a player makes more shots. For a marked spatial point process, this means that the mark and the intensity are associated. We propose a Bayesian joint model for the mark and the intensity of marked point processes, where the intensity is incorporated in the mark model as a covariate. Inferences are done with a Markov chain Monte Carlo algorithm. Two Bayesian model comparison criteria, the Deviance Information Criterion and the Logarithm of the Pseudo-Marginal Likelihood, were used to assess the model. The performances of the proposed methods were examined in extensive simulation studies. The proposed methods were applied to the shot charts of four players (Curry, Harden, Durant, and James) in the 2017–2018 regular season of the National Basketball Association to analyze their shot intensity in the field and the field goal percentage in detail. Application to the top 50 most frequent shooters in the season suggests that the field goal percentage and the shot intensity are positively associated for a majority of the players. The fitted parameters were used as inputs in a secondary analysis to cluster the players into different groups.

Keywords: MCMC; model selection; sports analytic.

1 Introduction

Shot charts are important summaries for basketball players. A shot chart is a spatial representation of the location and the result of each shot attempt by one player. Good defense strategies depend on good understandings of the offense players' tendencies to shoot and abilities to score. Reich et al. (2006) proposed hierarchical spatial models with spatially-varying covariates for shot attempt frequencies over a grid on the court and for shot success with shot locations fixed. Spatial point processes are commonly used to model random locations (e.g., Cressie 2015; Diggle 2013). Miller et al. (2014) used a low dimensional representation of related point processes to analyze shot attempt locations. Franks et al. (2015) combined spatial and spatio-temporal processes, matrix factorization techniques, and hierarchical regression models to analyze defensive skill. Many parametric models for spatial point process have been proposed in the literature, such as the Poisson process (Geyer 1999), the Gibbs process (Møller and Waagepetersen 2003), and the log Gaussian Cox process (LGCP) (Møller et al. 1998). When each point in a point process is companied with a random variable or vector known as mark, the resulting process is a marked point process (e.g., Banerjee, Carlin, and Gelfand 2014, Ch. 8). A shot chart can be modeled by a spatial marked point process with a binary mark showing the shot results.

The frequency of successful shots may be higher at locations where a player makes more shot attempts. This positive association is expected from different angles. More frequent shots suggests higher competence level and, hence, higher shooting accuracy. Higher accuracy also encourages more shooting since it means higher reward. In behavioral science, the matching law states that individuals will allocate their behavior according to the relative rates of reinforcement available for each option (Baum 1974; Staddon 1978). It predicts higher proportion of three-point shots taken relative to all shots to be associated with higher proportion of three-point shots scored relative to all shots scored (Alferink et al. 2009; Vollmer and Bourret 2000). The association might be more obvious for players who get fewer minutes and may be more selective to "prove their worth". It might be less so for players who have possession more often and may be less selective. Team strategies could affect the association in two opposite directions. Players with high three-point shot accuracy are more likely to be arranged in areas beyond the

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$$L(\boldsymbol{\Theta}|\mathbf{S},\mathbf{M}) \propto \prod_{i=1}^{N} \theta(\boldsymbol{s}_i)^{m(\boldsymbol{s}_i)} (1-\theta(\boldsymbol{s}_i))^{1-m(\boldsymbol{s}_i)} \times \left(\prod_{i=1}^{N} \lambda(s_i) \right) \exp\left(-\int_B \lambda(s) ds \right). \tag{3}$$

3.2 Prior specification

Vague priors are specified for model parameters. For λ_0, the gamma distribution is conjugate prior (e.g., Leininger et al. 2017). For $\boldsymbol{\beta}$, ξ, or $\boldsymbol{\alpha}$, there is no conjugate prior and we specify a vague, independent normal prior. In summary, we have

$$\begin{aligned} \lambda_0 &\sim G(a,b), \\ \boldsymbol{\beta} &\sim \mathrm{MVN}\left(\mathbf{0}, \sigma_\beta^2 \mathbf{I}_p\right), \\ \xi &\sim N\left(0, \sigma_\xi^2\right), \\ \boldsymbol{\alpha} &\sim \mathrm{MVN}\left(\mathbf{0}, \sigma_\alpha^2 \mathbf{I}_q\right), \end{aligned} \tag{4}$$

where $G(a, b)$ represents a Gamma distribution with shape a and rate b, respectively, $\mathrm{MVN}(\mathbf{0}, \boldsymbol{\Sigma})$ is a multivariate normal distribution with mean vector $\mathbf{0}$ and variance matrix $\boldsymbol{\Sigma}$, $(a, b, \sigma_\beta^2, \sigma_\xi^2, \sigma_\alpha^2)$ are hyper-parameters to be specified, and $\mathbf{I}_k$ is the k-dimensional identity matrix.

4 Bayesian computation

4.1 The MCMC sampling schemes

The posterior distribution of $\boldsymbol{\Theta}$ is

$$\pi(\boldsymbol{\Theta}|\mathbf{S},\mathbf{M}) \propto L(\boldsymbol{\Theta}|\mathbf{S},\mathbf{M}) \pi(\boldsymbol{\Theta}), \tag{5}$$

where $\pi(\boldsymbol{\Theta}) = \pi(\lambda_0)\pi(\boldsymbol{\beta})\pi(\xi)\pi(\boldsymbol{\alpha})$ is the joint prior density as specified in (4). In practice, we used vague priors with hyper-parameters $\sigma_\beta^2 = \sigma_\xi^2 = \sigma_\alpha^2 = 100$ and $a = b = 0.01$ in (4).

To sample from the posterior distribution of $\boldsymbol{\Theta}$ in (5), an Metropolis–Hasting within Gibbs algorithm is facilitated by R package nimble (de Valpine et al. 2017). The loglikelihood function of the joint model used in the MCMC iteration is directly defined using the `RW_llFunction()` sampler. The integration in the likelihood function (3) does not have a closed-form. It needs to be computed with a Riemann approximation by partitioning $\mathcal{B}$ into a grid with a sufficiently fine resolution. Within each grid box, the integrand $\lambda(\boldsymbol{s})$ is approximated by a constant. Then the integration of $\lambda(\boldsymbol{s})$ becomes a summation over all of the grid boxes.

4.2 Bayesian model comparison

To assess whether the intensity term is necessary in the mark model (2), model comparison criteria is needed. Within the Bayesian framework, DIC (Spiegelhalter et al. 2002) and LPML (LPML; Geisser and Eddy 1979; Gelfand and Dey 1994) are two well-known Bayesian criteria for model comparison. Using the method of Zhang et al. (2017), each criterion for the proposed joint model can be decomposed into one for the intensity model and one for the mark model conditioning on the point pattern for more insight on the model comparison.

The DIC for the joint model is

$$\begin{aligned} \mathrm{DIC} &= \mathrm{Dev}\left(\overline{\boldsymbol{\Theta}}|\mathbf{S},\mathbf{M}\right) + 2p_D, \\ p_D &= \overline{\mathrm{Dev}}(\boldsymbol{\Theta}|\mathbf{S},\mathbf{M}) - \mathrm{Dev}\left(\overline{\boldsymbol{\Theta}}|\mathbf{S},\mathbf{M}\right), \end{aligned} \tag{6}$$

where the deviance Dev is the negated loglikelihood function in Equation (3), $\overline{\mathrm{Dev}}$ is the mean of the deviance evaluated at each posterior draw of the parameters, $\overline{\boldsymbol{\Theta}}$ is the posterior mean of $\boldsymbol{\Theta}$, and p_D is known as the effective number of parameters. For the intensity model and the conditional mark model, the DIC can be computed with deviance, respectively,

$$\begin{aligned} \mathrm{Dev}_{\mathrm{intensity}}(\lambda_0, \boldsymbol{\beta}|\mathbf{S}) &= -2\left(\sum_{i=1}^{N} \log\lambda(\boldsymbol{s}_i) - \int_{\mathcal{B}} \lambda(\boldsymbol{s}) \mathrm{d}\boldsymbol{s} \right), \\ \mathrm{Dev}_{\mathrm{mark}}(\boldsymbol{\lambda}, \boldsymbol{\alpha}, \xi|\mathbf{M},\mathbf{S}) &= -2\sum_{i=1}^{N} \log f\left(m(\boldsymbol{s}_i)|\mathbf{S}; \lambda(\boldsymbol{s}_i), \boldsymbol{\alpha}, \xi, \mathbf{Z}(\boldsymbol{s}_i)\right), \end{aligned} \tag{7}$$

where $\boldsymbol{\lambda} = (\lambda(\boldsymbol{s}_1), \lambda(\boldsymbol{s}_2), \ldots, \lambda(\boldsymbol{s}_N))$, and $f(m(\boldsymbol{s}_i)|\lambda(\boldsymbol{s}_i), \boldsymbol{\alpha}, \xi, \mathbf{Z}(\boldsymbol{s}_i))$ is the conditional probability mass function of $m(\boldsymbol{s}_i)$ given $(\lambda(\boldsymbol{s}_i), \boldsymbol{\alpha}, \xi, \mathbf{Z}(\boldsymbol{s}_i))$. Clearly, the DIC for joint model is the summation of the DIC for the intensity model and the DIC for the conditional mark model. Models with smaller DIC are better models.

Calculation of the LPML for point process models is challenging because the usual conditional predictive ordinate (CPO) based on the leaving-one-out assessment is not applicable where the number of points N is random. Hu et al. (2019) recently suggested a Monte Carlo method to approximate the LPML for the intensity model as

$$\widehat{\mathrm{LPML}}_{\mathrm{intensity}} = \sum_{i=1}^{N} \log\tilde{\lambda}(\boldsymbol{s}_i) - \int_{\mathcal{B}} \overline{\lambda}(\boldsymbol{s}) \mathrm{d}\boldsymbol{s}, \tag{8}$$

where $\tilde{\lambda}(\boldsymbol{s}_i) = \left(\frac{1}{K} \sum_{k=1}^{K} \lambda^{(k)}(\boldsymbol{s}_i)^{-1} \right)^{-1}$, $\overline{\lambda}(\boldsymbol{s}) = \frac{1}{K} \sum_{k=1}^{K} \lambda^{(k)}(\boldsymbol{s})$, and $\{\lambda^{(k)}(\boldsymbol{s}_i) : k = 1, 2, \ldots, K\}$ is a posterior sample of size K of the parameters from the MCMC. The LPML for the conditional mark model can be calculated as usual (Chen, Shao, and Ibrahim 2000, Ch. 10). For the ith data point, define

$$\widehat{\mathrm{CPO}}_i^{-1} = \frac{1}{K}\sum_{b=1}^{K}\frac{1}{f\left(m(\boldsymbol{s}_i)|\lambda^{(K)}(\boldsymbol{s}_i),\boldsymbol{\alpha}^{(K)},\boldsymbol{\xi}^{(K)},\mathbf{Z}(\boldsymbol{s}_i)\right)},$$

where $\{\boldsymbol{\alpha}^{(K)},\boldsymbol{\xi}^{(K)}: k = 1,2,\ldots,K\}$ is a posterior sample of size K of the parameters from the MCMC. Then the LPML on mark model is

$$\widehat{\mathrm{LPML}}_{\mathrm{mark}} = \sum_{i=1}^{N}\log(\widehat{\mathrm{CPO}}_i). \tag{9}$$

The LPML for the joint model is then calculated as the sum of (8) and (9). Models with higher LPML are better models.

5 Simulation studies

To investigate the performance of the estimation, we generated data from a non-homogeneous Poisson point process defined on a square $\mathcal{B} = [-1,1]\times[-1,1]$ with intensity $\lambda(\boldsymbol{s}_i) = 100\lambda_0\exp(\beta_1 x_i + \beta_2 y_i)$, where $\boldsymbol{s}_i = (x_i, y_i)\in\mathcal{B}$ is the location for every data point. For each $\boldsymbol{s}_i$, $i = 1,\ldots,N$, the mark $m(\boldsymbol{s}_i)$ follows a logistic model with two covariates in addition to λ and intercept:

$$\begin{aligned} m(\boldsymbol{s}_i) &\sim \mathrm{Bern}(p_i),\\ \mathrm{logit}(p_i) &= \xi\lambda(\boldsymbol{s}_i) + \alpha_0 + \alpha_1 Z_{1i} + \alpha_2 Z_{2i}. \end{aligned} \tag{10}$$

The parameters of the model were designed to give point counts that are comparable to the basketball shot chart data. We fixed $(\beta_1,\beta_2) = (2,1)$, $\xi = 0.5$, $\alpha_0 = 0.5$, and $\alpha_2 = 1$. Three levels of α_1 were considered, $\alpha_1\in\{0.8,1,2\}$, in order to compare the performance of the estimation procedure under different magnitudes of the coefficients in the mark model. Two levels of λ_0 were considered, $\lambda_0\in\{0.5,1\}$, which controls the mean of the number of points on $\mathcal{B}$. It is easy to integrate in this case the intensity function over $\mathcal{B}$ to get the average number of points being 850 and 1700, respectively, for $\lambda_0 = 0.5$ and 1. The numbers are approximately in the range of the NBA basketball shot charts in Section 2. In the mark model, covariate Z_1 was generated from the standard normal distribution; two types of Z_2 were considered, standard normal distribution or Bernoulli with rate 0.5. The resulting range of the Bernoulli rate of the marks was within (0.55, 0.78) for all the scenarios.

For each setting, 200 data sets were generated. R package `spatstat` (Baddeley et al. 2005) was used to generate the Poisson point process data with the given intensity function. The priors for the model parameters were set to be (4) with the hyper-parameters $\sigma_\beta^2 = \sigma_\xi^2 = \sigma_\alpha^2 = 100$ and $a = b = 0.01$. The grid used to calculate the integration in likelihood function had resolution 100×100. For each data set, a MCMC was run for 20,000 iterations with the first 10,000 treated as burn-in period. For each parameter, the posterior mean was used as the point estimate and the 95% credible interval was constructed with the 2.5% lower and upper quantiles of the posterior sample.

Tables 3 and 4 in Appendix summarize the simulation results for the scenarios of standard normal Z_2 and Bernoulli Z_2, respectively. The empirical bias for all the settings are close to zero. The average posterior standard deviation from the 200 replicates is very close to the empirical standard deviation of the 200 point estimates for all the parameters, suggesting that the uncertainty of the estimator are estimated well. Consequently, the empirical coverage rates of the credible intervals are close to the nominal level 0.95. As α_1 increases, the variation increases in the mark parameter estimates but does not change in the intensity parameter estimates. As λ_0 increases, the variations of the estimates for both intensity and mark parameters get lower. Between the continuous and binary cases of Z_2, the variation in the estimates is higher in the latter case, especially for the coefficient of Z_2.

6 NBA players shot chart analysis

6.1 Covariates construction

To capture the shot styles of individual players in their shot intensity model, we follow Miller et al. (2014) to construct basis covariates that are interpreted as archetypal intensities or "shot types" used by the players. The focus is on the 35 ft by 50 ft rectangle on the side of the backboard in the offensive half court. The origin of the Cartesian coordinates (x, y) is replaced at the bottom left corner so that $x\in[0,50]$ and $y\in[0,35]$. The rectangle was evenly partitioned into 50×35 grid boxes of 1 ft by 1 ft. Our bases construction is slightly different from that of Miller et al. (2014) in the preparation for the Nonnegative matrix factorization (NMF). First, we used a kernel estimation instead of an LGCP model to estimate the 50×35 intensity matrix of each individual players, which is easier to compute and more accurate in the sense of intensity fitting accuracy. Second, we used historical data instead of the current season data. In particular, a kernel estimate of the 50×35 intensity matrix for each of the 407 players in the previous season (2016–2017) who had made over 50 shots was used as input for the NMF. As in Miller et al. (2014), we obtained 10 bases using R package NMF (Gaujoux and Seoighe 2010).

Figure 2 displays the 10 nonnegative matrix bases that can be used as covariates for the intensity matrix fitting.

Figure 2: Intensity matrix bases heat plots.

They are similar to those in the literature (Franks et al. 2015; Miller et al. 2014). Each basis is nicely interpreted as a certain shot type. For example, basis 1 is long two-points, bases 2–3 are left/right wing threes, bases 4–5 are left/right/center restricted area two-points, basis 7 is top of key threes, basis 8 is center threes, basis 9 is corner threes, and basis 10 is mid-range twos. When used as covariates in modeling individual shot intensity, their coefficients characterize the shooting style of each player.

The influence of intensity on shot accuracy might be different for different shot type. Players' shot selection may be biased towards three point shot for higher reward (Alferink et al. 2009; Skinner and Goldman 2015). This can result in higher intensity for three-point shot at locations with not high accuracy. To capture this tendency, an interaction term between the intensity and the shot type is introduced to the mark model. In addition to intensity and interaction term between intensity and shot type, other covariates in the mark model include distance to the basket and non-spatial covariates such as seconds left to the end of the period, dummy variables for five different periods with first period as reference, and the indicator of opponent made to the playoff in the last season.

6.2 Model comparison

The joint model (1)–(2) was fitted for each player with the hyper-parameters in (4) set as $\sigma_\beta^2 = \sigma_\xi^2 = \sigma_\alpha^2 = 100$ and $a = b = 0.01$. The numerical integration in evaluating the joint log-likelihood (3) was based on the same 50×35 grid as that used in constructing the basis shot styles from NMF. To check the importance of intensity as covariate in the mark component, we also fitted the model with the restriction $\xi = 0$. For each model fitting, 60,000 MCMC iterations were run. The first 20,000 were discarded as the burn-in period and the rest were thinned by 10, which led to an MCMC sample of size 4000. The trace plots of the MCMC

were checked and the convergence of all the parameters were confirmed. The reported results were obtained from a second run after insignificant covariates were removed to avoid possible collinearity among some variables; for example, basis 6 (restricted area two-points) appears to be well approximated by a combination of basis 4 (left restricted area two-points) and basis 5 (right restricted area two-points).

Table 1 summarizes the DIC and LPML for the full joint model and its two components. The smallest absolute difference is 8.6 in DIC and 4.2 in LPML for Durant; the largest absolute difference is 41.2 in DIC and 20.4 in LPML for James. The DIC has a rule of thumb similar to AIC in decision making (Spiegelhalter et al. 2002, p. 613): a difference larger than 10 is substantial and a difference about 2–3 does not give an evidence to support one model over the other. For LPML, a difference less than 0.5 is "not worth more than to mention" and larger than 4.5 can be considered "very strong" (Kass and Raftery 1995). With these guidelines applied to DIC and LPML, the mark model with shot intensity included as a covariate has a clear advantage relative to the model without it for Durant, Harden, and James, but not for Curry, an interesting result which will be discussed in the next subsection. The difference in DIC and LPML between the models with and without $\xi = 0$ comes from the mark component. The two criteria for the intensity component are almost the same with and without $\xi = 0$. This is expected because the marks may contain little information about the intensities, and intensity fitting results are not influenced by the mark model significantly.

In order to have a direct comparison of improvement of mark model by using the preferred model, we calculate the mean squared error (MSE) of fitted mark models with and without intensity as a covariate. The preferred models for all four players, which are intensity independent model for Curry and intensity dependent model for other three players, can reduce the MSE by 2.7, 1.3, 2.0, and 7.0%.

6.3 Fitted results

Table 2 summarizes the posterior mean, posterior standard deviation, and the 95% highest posterior density (HPD) credible intervals for the regression coefficients in the models for Curry, Durant, Harden, and James as selected by the DIC and LPML. Only significant covariates are displayed as determined by whether or not the 95% HPD credible intervals cover zero in the first run. The reported results were from the second run after insignificant covariates were removed.

The coefficients of the 10 basis shot styles in the intensity model describe the composition of each individual player's shot style. After being exponentiated, they represent a multiplicative effect on the baseline intensity. So they are comparable across players as a relative scale. The four players are quite different in the coefficients of a few well-interpreted bases. Curry's rate of corner threes was the highest among the four. Durant has the least rate of corner threes and highest rate of long/mid-range two-pointers. Harden had the least rate of long two-pointers and highest rate of top of key threes. Curry and Harden had less two point shots but more three point shots than Durant and James. James seemed to prefer to shot on the left side of court for three point shots more than the other three players. All four had high rate of center threes. Figure 3 (upper) shows the fitted intensity surfaces of the four players. These results echo the findings in earlier works (Franks et al. 2015; Miller et al. 2014).

Table 1: Summaries of DIC and LPML for the models for Curry, Durant, Harden, and James with and without $\xi = 0$.

			Curry	Durant	Harden	James
Joint model	DIC	$\xi \neq 0$	2391.3	2977.1	1744.4	760.8
		$\xi = 0$	2379.2	2985.7	1753.7	802.0
	LPML	$\xi \neq 0$	−1195.7	−1489.6	−872.4	−380.8
		$\xi = 0$	−1189.6	−1493.8	−877.0	−401.2
Intensity	DIC	$\xi \neq 0$	1352.8	1593.0	12.3	−1012.9
		$\xi = 0$	1352.5	1593.0	12.2	−1013.4
	LPML	$\xi \neq 0$	−676.5	−797.4	−6.2	506.3
		$\xi = 0$	−676.3	−797.4	−6.2	506.5
Mark	DIC	$\xi \neq 0$	1038.4	1384.1	1732.1	1773.7
		$\xi = 0$	1026.7	1392.8	1741.5	1815.4
	LPML	$\xi \neq 0$	−519.2	−692.2	−866.2	−887.1
		$\xi = 0$	−513.3	−796.4	−870.7	−907.7

Table 2: Estimated coefficients in the joint models for Curry, Durant, Harden, and James.

Player	Model	Covariates	Posterior mean	Posterior SD	95% Credible Interval
Curry	Intensity	Baseline (λ_0)	0.236	0.012	(0.213, 0.261)
		Basis 1 (long two-pointers)	0.248	0.041	(0.167, 0.328)
		Basis 2 (right wing threes)	0.290	0.025	(0.236, 0.335)
		Basis 3 (left wing threes)	0.190	0.028	(0.132, 0.243)
		Basis 4 (left restricted area)	0.185	0.017	(0.152, 0.217)
		Basis 7 (top of key threes)	0.141	0.026	(0.091, 0.193)
		Basis 8 (center threes)	0.636	0.037	(0.563, 0.708)
		Basis 9 (corner threes)	0.121	0.019	(0.085, 0.158)
	Mark	Intercept	−0.165	0.092	(−0.338, 0.022)
		Distance	−0.270	0.064	(−0.396, −0.145)
Durant	Intensity	Baseline (λ_0)	0.372	0.015	(0.342, 0.401)
		Basis 1 (long two-pointers)	0.465	0.039	(0.386, 0.539)
		Basis 2 (right wing threes)	0.219	0.028	(0.163, 0.270)
		Basis 3 (left wing threes)	0.097	0.032	(0.036, 0.162)
		Basis 4 (left restricted area)	0.149	0.028	(0.096, 0.206)
		Basis 6 (restricted area)	−0.107	0.027	(−0.160, −0.056)
		Basis 7 (top of key threes)	0.071	0.027	(0.014, 0.121)
		Basis 8 (center threes)	0.634	0.038	(0.562, 0.707)
		Basis 9 (corner threes)	−0.074	0.036	(−0.147, −0.007)
		Basis 10 (mid-range twos)	0.479	0.036	(0.408, 0.550)
	Mark	Intercept	−0.353	0.127	(−0.609, −0.114)
		Intensity (λ)	1.237	0.430	(0.393, 2.065)
		Distance	−0.351	0.068	(−0.481, −0.209)
Harden	Intensity	Baseline (λ_0)	0.348	0.015	(0.319, 0.378)
		Basis 1 (long two-pointers)	−0.169	0.045	(−0.258, −0.085)
		Basis 2 (right wing threes)	0.193	0.021	(0.154, 0.236)
		Basis 3 (left wing threes)	0.084	0.025	(0.038, 0.135)
		Basis 4 (left restricted area)	0.235	0.023	(0.186, 0.277)
		Basis 6 (restricted area)	0.127	0.022	(0.085, 0.172)
		Basis 7 (top of key threes)	0.247	0.018	(0.209, 0.281)
		Basis 8 (center threes)	0.657	0.029	(0.598, 0.712)
		Basis 10 (mid-range twos)	0.086	0.023	(0.043, 0.133)
	Mark	Intercept	−0.453	0.067	(−0.582, −0.323)
		Intensity (λ)	1.291	0.201	(0.903, 1.686)
James	Intensity	Baseline (λ_0)	0.423	0.016	(0.395, 0.457)
		Basis 1 (long two-pointers)	0.113	0.035	(0.045, 0.181)
		Basis 3 (left wing threes)	0.165	0.028	(0.114, 0.223)
		Basis 4 (left restricted area)	0.166	0.019	(0.128, 0.204)
		Basis 6 (restricted area)	0.130	0.016	(0.098, 0.161)
		Basis 7 (top of key threes)	0.087	0.026	(0.037, 0.136)
		Basis 8 (center threes)	0.544	0.031	(0.483, 0.603)
		Basis 9 (corner threes)	0.069	0.023	(0.023, 0.111)
		Basis 10 (mid-range twos)	0.246	0.025	(0.200, 0.296)
	Mark	Intercept	−0.447	0.073	(−0.588, −0.304)
		Intensity (λ)	0.632	0.115	(0.418, 0.861)
		Distance	−0.326	0.056	(−0.433, −0.207)

The results from the mark model conditioning on the intensity are the major contribution of this work. All non-spatial covariates were insignificant and were dropped from the model, except shot distance. The coefficient of the intensity was found to be significantly positive for Durant, Harden, and James, but not for Curry. That is, for the players excluding Curry, shot accuracy was higher where they shot more frequently. The interaction between the intensity of shot type (two- vs. three-point) was not significant for any player, suggesting that, for those whose shot frequency and shot accuracy were positively associated, the association was not influenced directly by shot

Figure 3: Fitted shot intensity surfaces (upper) and expected score surfaces (lower) of Curry, Durant, Harden and James on the same scale. Redder means higher.

rewards. The magnitude of coefficient of the intensity shows how strong this dependence is. The association is much weaker (about a half) for James compared to Durant and Harden. Shot distance was found to have a significantly negative effect on shot accuracy for Curry, Durant, and James, but not for Harden. The presence of both shot distance and intensity in the shot accuracy model means that among locations with the same accuracy but different rewards (two- vs. three-point), three-point locations tend to have higher intensities. This reflects the bias of shooting intensity to three-point shot due to higher rewards (Alferink et al. 2009). Since shot distance was not significant in Harden's model, he could make more three-point shots for higher rewards.

Curry's mark model only included a single covariate shot distance with a significantly negative coefficient. At shot locations with the same shot distance, Curry's shot accuracy was not affected by his shot frequency, which makes him hard to guard against for a defense team. From an alternative direction of reasoning, Curry's results suggest that he did not shoot more often at locations where his shot accuracy was higher, which might not be optimal from the team strategy point of view. The might be due to his injury in that season and reduced time on court. He could

make more shots where his accuracy is higher to improve scoring efficiency.

The fitted mark model allows combining shot accuracy and shot frequency to construct an expected score map for each player; see Figure 3 (lower). This plot is more informative than a shooting accuracy plot because the latter would contain no value at locations where there were few or no shots. Curry had a more obvious scoring pattern of corner threes among the four. Durant and James had more two point scores and less three point scores than Curry and Harden. Curry and Harden's two point scores were more concentrated in the restricted area than Durant and James.

To get an idea about the intensity dependent effect on shot accuracy averaged over top players, we analyzed all shots attempted by the top 20 most frequent shooters, which cover Harden and James, but not Curry and Durant. The 20 players' data were pooled and treated as one virtual player. Due to computational feasibility, we could not include more players in the pool. The fitted coefficient of the intensity divided by 20 gives an "elite average" of the intensity dependent effect, which is 1.023. Compared with the results in Table 2, Harden and Durant's fitted coefficients were above the average, while James and Curry's were below the average (Curry's fitted coefficient can be treated as 0 since his result favors intensity independent model). The ranking relative to the elite average could be a measure in assessing the players' efficiency in shot location selection. Players with a fitted coefficient below the elite average might have room to improve their score efficiency through shot selection.

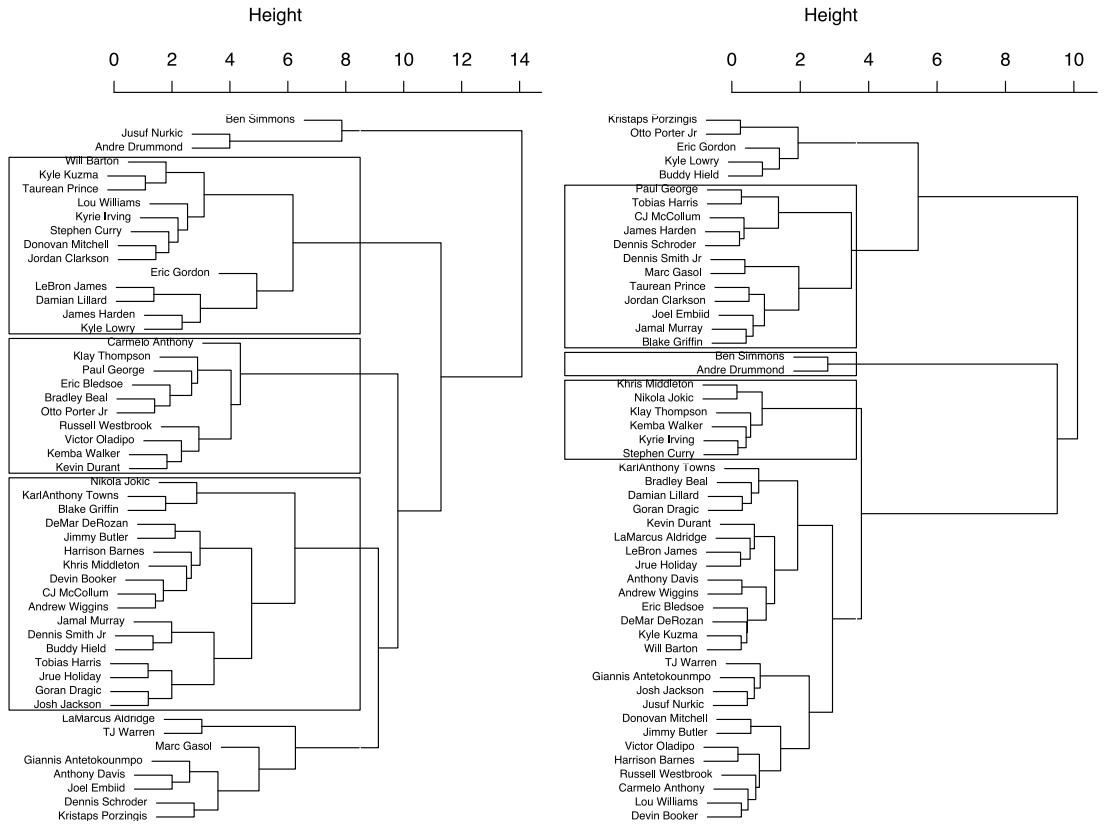
6.4 Application to top 50 most frequent shooters

We further applied the same analysis to each of the top 50 most frequent shooters in the 2017–2018 regular season. The number of shots of the 50 players ranged from 813 (Andre Drummond) to 1,517 (Russell Westbrook). Among them, 40 players' data favored the intensity dependent model ($\xi \neq 0$) in terms of DIC and LPML. Their estimated coefficients of the intensity in the mark model were all positive; the interactions between the intensity and the shot type (two- vs. three-point) were all not significantly different from zero. That is, 80% of the most frequent shooters in that season had positive association between shot intensity and shot accuracy, and the association did not vary with shot rewards. For the 10 players who had intensity independent mark models similar to Curry, shot distance was found to be significantly negative in every model.

The estimated coefficients in the joint model can be used as features to cluster the players into groups. With Curry added in, estimates from a total of 51 players were used as inputs to a cluster analysis. Both clustering for shot patterns based on the estimated coefficients in the intensity model and clustering for the accuracy–intensity relationship based on the mark model given intensity were considered. For the shot pattern clustering, only 10 coefficients of the basis styles, with the baseline intensity excluded, were used to focus on the distribution of the pattern instead of the total count of shots. The clustering of the accuracy–intensity relationship clustering only used the coefficients of intensity and shot distance in addition to the intercept because the other coefficients were found to be insignificant for most of the players. We used the hierarchical clustering method using the minimum variance criterion of Ward (1963) as implemented in R (Murtagh and Legendre 2014).

Figure 4a displays the results of clustering the 51 players by their shot patterns into five groups. The first group only contains three players who made mostly two-point shots. The second group includes, interestingly, Curry, Harden, and James. The closest players to Curry, Durant, and James were, respectively, Kyrie Irving, Kyle Lowry, and Damian Lillard. Players in this group had relative small coefficients for bases 1 and 10, and large coefficients for bases 3 and 8. That is, they had less long/mid-range twos and more threes, especially left wing threes. Players in the third group, which includes Durant, had large coefficients for bases 1 and 10, showing that they had more long/mid-range two-pointers. The closest player to Durant was Kemba Walker. Group four includes players with small coefficient for basis 6 and large coefficient for basis 9, which means that they had less two-pointers from the restricted area and more corner threes. The last group contains to players with small coefficients for bases 3 and 9, and large coefficient for basis 10, indicating less left wing threes and corner threes, but more mid-range twos.

The clustering results of the 51 players by the characteristics of their shot accuracy in relation to their shot intensity are shown in Figure 4b. Group two has Harden and other players whose mark model contained the shot intensity but not distance. Group four, which includes Curry, contains half of the players whose shot intensity was insignificant in their mark model. Group five is the largest group, which includes Durant and James. The players in this group had significant shot distance effect on their accuracy. Most of them had intensity in the mark model with a relatively small coefficients, and five of them had



(a) Intensity Model **(b)** Mark Model

Figure 4: Hierarchical clustering of 51 NBA players into five groups based on fitted coefficients in the intensity model and the mark model.

intensity insignificant. The first group includes players with intensity but not distance in the mark model, which is similar to Group two, but the magnitude of the coefficient for the intensity was the largest among all the players, suggesting the strongest dependence between shot intensity and shot accuracy. Players here were more likely to shoot at locations with higher accuracy rates. The third group has only two players, Simmons and Drummond, whose coefficients for shot distance were much larger than others' in magnitude, which was expected because the two players shot mostly in the restricted area.

7 Discussion

We proposed a Bayesian marked spatial point process to model both the shot locations and shot outcomes in NBA players' shot charts. Basis shot styles constructed from the NMF method (Miller et al. 2014) were included as covariates in the intensity for the Poisson point process model and the logistic model for shot outcomes. For a majority of the top players, a positive association between the shot intensity and shot accuracy was reported. The association did not vary significantly according to the shot rewards. Players whose shot intensity was not found to affect their shot accuracy (e.g., Curry) may be hard to defend against. From the offense perspective, these players could score more by making more shots where they shot more frequently. The cluster analyses based on the fitted coefficients characterizing the shot pattern and shot accuracy are quite unique. Unlike other cluster analyses, (e.g., Zhang et al. 2018), the data input here are not directly observed but estimated from fitting a model to the shot charts. Consequently, less obvious insights could be discovered.

A few directions of further work are worth investigating. Our proposed model is univariate in the sense that each player is modeled separately. A full hierarchical model for pooled data from multiple players in one season may be useful with a random effect at the player level for certain parameters. The number of basis shot styles was set to 10 as suggested by Miller et al. (2014). It would be interesting to find an optimal number of basis through model comparison criteria like DIC and LPML. An important factor for shot accuracy is the shot clock time

remaining (Skinner 2012), but it is not available in the dataset we obtained. It should be added to the mark model if available. Our spatial Poisson process model formulates a linear relationship between the spatial covariates and the log intensity, which cannot capture more complicated spatial trend of the intensity of spatial point pattern. Including some Bayesian non-parametric methods like finite mixture model (Miller and Harrison 2018) may help increase the accuracy of the estimation of spatial point pattern.

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Appendix

This section shows the tables of simulation results.

Table 3: Summaries of the bias, standard deviation (SD), average of the Bayesian SD estimate ($\widehat{\text{SD}}$), and coverage rate (CR) of 95% credible intervals when Z_2 is continuous: $\xi = \alpha_0 = 0.5$, $\alpha_2 = 1$, $(\beta_1, \beta_2) = (2, 1)$ and $Z_2 \sim N(0, 1)$.

			$\lambda_0 = 0.5$				$\lambda_0 = 1$			
α_1	Model	Para	Bias	SD	$\widehat{SD}$	CR	Bias	SD	$\widehat{SD}$	CR
0.8	Intensity	λ_0	0.01	0.04	0.04	0.96	0.01	0.06	0.06	0.93
		β_1	−0.06	0.11	0.11	0.90	−0.06	0.09	0.08	0.88
		β_2	−0.05	0.09	0.09	0.94	−0.03	0.06	0.06	0.92
	Mark	ξ	0.11	0.57	0.60	0.97	0.04	0.22	0.22	0.96
		α_0	0.01	0.20	0.20	0.95	0.00	0.14	0.14	0.97
		α_1	0.03	0.13	0.13	0.94	0.01	0.09	0.09	0.94
		α_2	0.03	0.14	0.14	0.95	0.01	0.10	0.10	0.93
1	Intensity	λ_0	0.00	0.04	0.04	0.94	0.00	0.06	0.06	0.94
		β_1	−0.05	0.11	0.11	0.94	−0.05	0.08	0.08	0.91
		β_2	−0.03	0.10	0.09	0.92	−0.04	0.07	0.06	0.92
	Mark	ξ	0.03	0.60	0.61	0.95	0.05	0.21	0.22	0.96
		α_0	0.00	0.20	0.20	0.95	−0.01	0.14	0.14	0.95
		α_1	0.01	0.13	0.14	0.97	0.01	0.10	0.10	0.96
		α_2	0.03	0.14	0.14	0.95	0.01	0.10	0.10	0.94
2	Intensity	λ_0	0.00	0.04	0.04	0.94	0.00	0.06	0.06	0.95
		β_1	−0.06	0.12	0.11	0.91	−0.04	0.08	0.08	0.93
		β_2	−0.03	0.09	0.09	0.94	−0.03	0.07	0.06	0.91
	Mark	ξ	0.04	0.71	0.69	0.94	0.05	0.23	0.24	0.96
		α_0	0.02	0.22	0.23	0.95	−0.01	0.15	0.16	0.97
		α_1	0.08	0.23	0.21	0.93	0.03	0.15	0.15	0.94
		α_2	0.03	0.17	0.16	0.93	0.02	0.11	0.11	0.95

Table 4: Summaries of the bias, standard deviation (SD), average of the Bayesian SD estimate ($\widehat{\text{SD}}$), and coverage rate (CR) of 95% credible intervals when Z_2 is binary: $\xi = \alpha_0 = 0.5$, $\alpha_2 = 1$, $(\beta_1, \beta_2) = (2, 1)$ and $Z_2 \sim \text{Bernoulli}(0.5)$.

			$\lambda_0 = 0.5$				$\lambda_0 = 1$			
α_1	Model	Para	Bias	SD	$\widehat{SD}$	CR	Bias	SD	$\widehat{SD}$	CR
0.8	Intensity	λ_0	0.00	0.04	0.04	0.94	0.01	0.06	0.06	0.95
		β_1	−0.05	0.12	0.11	0.88	−0.05	0.08	0.08	0.90
		β_2	−0.02	0.10	0.09	0.94	−0.03	0.06	0.06	0.95
	Mark	ξ	0.07	0.62	0.61	0.94	0.04	0.20	0.23	0.96
		α_0	0.00	0.23	0.22	0.93	0.01	0.16	0.16	0.95
		α_1	0.03	0.13	0.13	0.96	0.01	0.10	0.10	0.94
		α_2	0.03	0.25	0.24	0.96	0.01	0.19	0.18	0.95
1	Intensity	λ_0	0.00	0.04	0.04	0.94	0.00	0.06	0.06	0.94
		β_1	−0.06	0.11	0.11	0.93	−0.04	0.09	0.08	0.89
		β_2	−0.04	0.08	0.09	0.94	−0.02	0.06	0.06	0.93
	Mark	ξ	0.10	0.64	0.63	0.94	0.09	0.22	0.23	0.94

Table 4: (continued)

			$\lambda_0 = 0.5$				$\lambda_0 = 1$			
α_1	**Model**	**Para**	**Bias**	**SD**	$\widehat{SD}$	**CR**	**Bias**	**SD**	$\widehat{SD}$	**CR**
		α_0	0.01	0.23	0.23	0.97	−0.03	0.16	0.16	0.94
		α_1	0.03	0.15	0.14	0.92	0.01	0.11	0.10	0.92
		α_2	0.02	0.27	0.25	0.93	0.02	0.17	0.18	0.96
2	Intensity	λ_0	0.00	0.04	0.04	0.95	0.01	0.06	0.06	0.94
		β_1	−0.05	0.11	0.11	0.92	−0.05	0.08	0.08	0.90
		β_2	−0.04	0.09	0.09	0.91	−0.04	0.06	0.06	0.92
	Mark	ξ	0.06	0.73	0.70	0.94	0.07	0.28	0.25	0.93
		α_0	0.03	0.29	0.26	0.93	−0.01	0.20	0.19	0.94
		α_1	0.06	0.21	0.21	0.94	0.05	0.15	0.15	0.94
		α_2	0.03	0.31	0.28	0.93	0.03	0.19	0.20	0.94

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